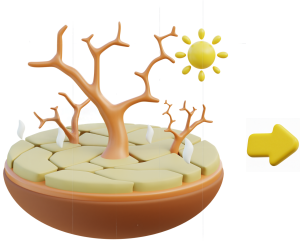
Oasis in the Predictability Desert?

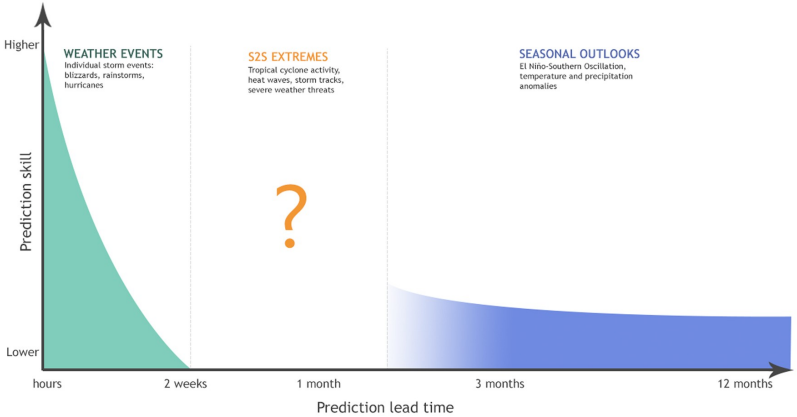


2–6 weeks ahead

The predictability desert

Why Is This So Hard to Predict?

The S2S Prediction Gap



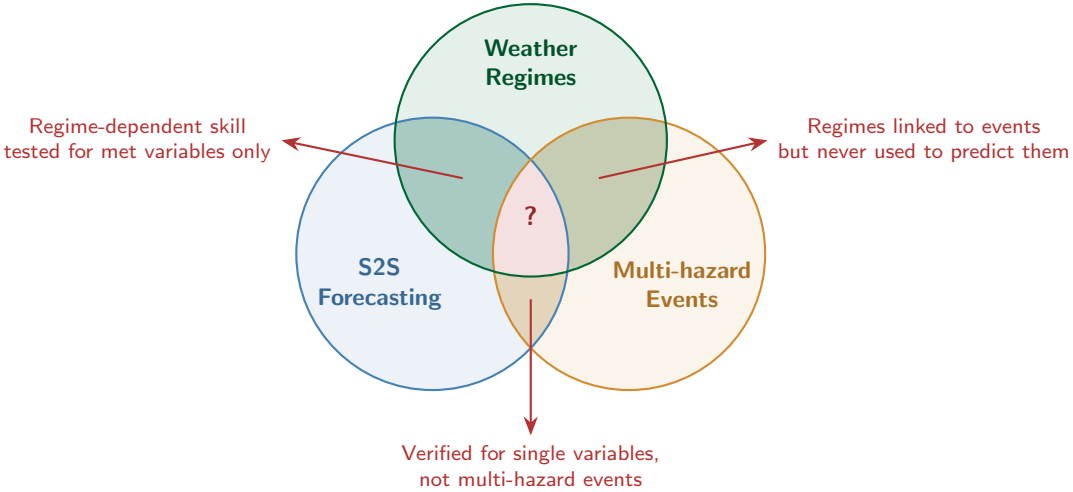
Source: Mariotti et al. (2018)

24%

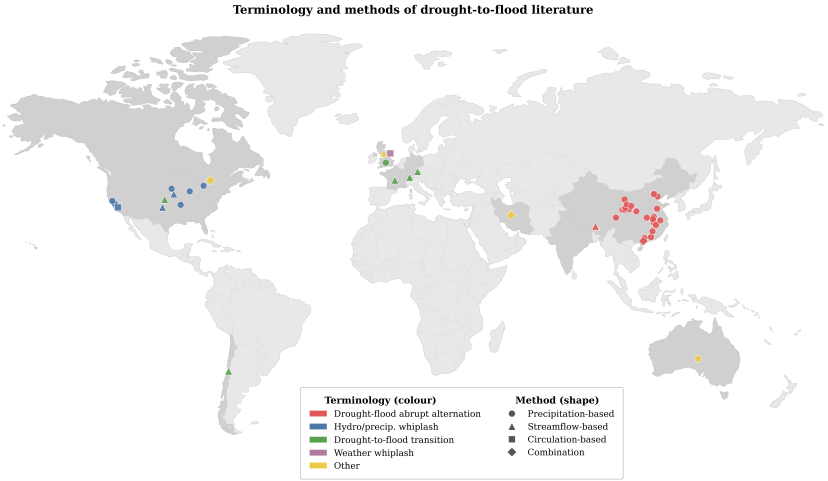
of all floods globally follow drought

Matanó et al. (2024)

What Does the Literature Say?



How Do You Detect These Events?



Joint indices

Compress the swing into one number

vs

Episode-linking

Detect separately, link if close in time

What Am I Testing?

1. *How does prediction skill at S2S lead times compare across forecast variables, impact variables, and multi-hazard events?*
2. *Under what atmospheric conditions does this skill emerge?*

Met skill → Impact skill → Event skill → Regime conditioning → ML

Paper 1
Drought-to-flood

Paper 2
Energy source droughts

Paper 3
ANTICIPATE

Where Does ML Fit?

ECMWF forecast → **Bias correction** → **Hydro model** → **Streamflow** → **Event detection**

Post-process
Prior to hydro model

Augment
LSTMs for rainfall-runoff

Bypass
Forecast → streamflow directly via ML

Regime conditioning
Atmospheric state as input to any ML model

Where Am I On This?

Subseasonal (EEFH)

Newer physics
183 inits / year
11 members

↑ **Temporal sampling**

vs

Seasonal (SEAS5)

Older physics
12 inits / year
25 members

↑ **Ensemble sampling**

2006–2016 | 0.3° grid | t2m, tp, ws10

Deterministic verification

Station and spatial results

Conclusions

1. Nobody has tested whether S2S forecast skill survives to multi-hazard events

2. Pipeline: met skill $\rightarrow$ impact skill $\rightarrow$ event skill

3. ML across the pipeline to boost event skill